

Carl J. Factora

Software Engineer
ivanthetricourne.io

EXPERIENCE

Bloomberg L.P. New York, NY
Senior Software Engineer May 2024-Present

Foamspace Corp Brooklyn, NY
Software Engineer Aug 2019-May 2024

- Managed and deployed a full-stack application to oversee a decentralized proof-of-location protocol
 - Designed an in-house TCP/TLS interface to efficiently manage and communicate with a globally deployed fleet of IoT radio devices (FOAM nodes) used for terrestrial localization
 - Designed a data pipeline utilizing Haskell's conduit library capable of asynchronously processing large amounts of localization data as part of a decentralized alternative to GPS
 - Collaborated with consulting teams to design a web app interface in PureScript used by FOAM zone operators for testing and improving localization performance
- Contributed to open-source Haskell and PureScript libraries leveraged by the blockchain and web3 community
 - Secured an open source grant for completing a Haskell library enabling the development of blockchain apps interfacing with the Cosmos SDK
- Assumed a mixed-role encompassing backend, frontend, and dev-ops responsibilities supporting a small engineering team (<4 software engineers) to meet operational needs

Droit Financial Tech., LLC New York, NY
Software Developer Aug 2017-July 2019

- Spearheaded the design and maintenance of a low-latency decision engine API used at some of the world's largest banks for financial regulatory compliance that remains the core of the company's primary product
- Designed a domain-specific language (DSL) and its associated tooling; improved API request latency by >40%

EDUCATION

The Recurse Center New York, NY
Hacker School Feb 2017-June 2017

- Designed and authored an online, interactive book for functional programming inspired by Eloquent JavaScript
- Self-taught various topics of interest, focusing on type-driven development and web development

Indiana University, Bloomington Bloomington, IN
Bachelor's Degree in Computer Science and English Literature 2012-2016

- Instructed numerous graduate and undergraduate courses on programming language theory
- Contributed to Essentials of Compilation: An Incremental Approach in Racket by Jeremy G. Siek
- Conducted research with Daniel P. Friedman on dependent type theory and constructive proofs
- Associate instructor for Advanced Functional Programming courses following graduation

ADDITIONAL INFORMATION

- **Programming Languages:** OCaml, Haskell, PureScript, Clojure, Elm, Racket/Scheme, JavaScript
- **Technologies:** Docker, PostgreSQL, Redis, Kubernetes